

Consolidated VLA results (HT campaign) — single source for the paper tables

Detail files: vla_results.md (GR1 curves/per-task, pi0.5 per-suite), multimodality_table.md, memory notes nu-is-inert-heteroscedasticity-is-the-method.md. Updated 2026-08-27.

A. Cross-stack headline — single-pass HT vs flow / L1 / MSE (budget-matched, all entries annotated @steps)

Blank = not run. HT entries carry their nu. Within a row every number is at the same eval protocol; "@steps" is the training checkpoint evaluated.

stack (chunk d)	protocol	flow	L1	MSE	HT incumbent (nu = 2d)	HT swept- best	HT fixed c = 2 (nu = 4d)
GR00T GR1, robocasa (232)	24 tasks x 20 eps, official harness	44.1 @60k (pub 44.5 @60k)	17.2 @60k	37.8 @60k	nu=464: 47.5 @60k	nu=928: 51.5 @60k (nu=2: 24.5; nu=0.5: 44.8; nu=464: 47.5)	nu=928: 51.5 @60k
GR00T WidowX, bridge (56)	7 tasks x 50 eps (effective 10 scenes/task)	57.1 @20k			nu=112: 62.9 @20k	nu=224: 67.4 @20k (flat 49-67 over nu=2..448)	nu=224: 67.4 @20k
GR00T Google Robot, fractal (56)	6 tasks x 100 eps, SimplerEnv	67.3 @20k					nu=224: 63.5 @20k
OpenVLA- OFT libero-long (56)	10 tasks x 10 eps (released ref: 500 eps, best ckpt)		released 94.5 (best ckpt <=150k)		nu=112: 82 @120k (arm endpoint)	nu=1024: 93.0 @150k (465/500; a 100-ep screen read 98)	nu=224: 94.0 @150k (470/500)
OpenVLA- OFT libero- spatial	10 tasks x 50 eps (500)		released 97.6		nu=112: 98.4 @50k (492/500)		nu=224: 97.4 @150k (487/500)
OpenVLA- OFT libero-goal	same		released 97.9		nu=112: 98 @90k (97.6 @30k, 500 eps)	nu=1024: 97.6 @150k (488/500); 96.0 @50k (480/500)	nu=224: 96.6 @150k (483/500); 96.8 @50k (484/500, their protocol point)
OpenVLA- OFT libero- object	same		released 98.4		nu=112: 95 (best ckpt)	nu=1024: 95.8 @20k (479/500; a 100-ep	nu=224: 95.6 @150k (478/500); 96.6

stack (chunk d)	protocol	flow	L1	MSE	HT incumbent (nu = 2d)	HT swept- best	HT fixed c = 2 (nu = 4d)
						screen read 99)	@100k (483/500, ckpt selected on a 100-ep sweep and confirmed at 500)
pi0.5 LIBERO (350), 4- suite avg	4 x 1000 eps	published ~96.9		96.8 @30k	nu=700: 97.6 @30k (spatial 98.1 / goal 97.6 / object 99.8 / long 94.7)	(nu=2 97.0 @30k was a 1-GPU run, 8x fewer samples: not comparable)	nu=1400: 96.6 @30k (97.4 / 97.1 / 99.4 / 92.6)
Cosmos3 libero-10 (160)	10 tasks x 10 eps, 30- step sampler	96 @2000			nu=320: 93 @2000	nu=512: 96 @2000	nu=640: 96 @2000

Reading across rows: single-pass HT matches or beats its iterative baseline on every stack once nu is at a light-but-finite operating point (GR1 51.5 vs 44.1; WidowX 62.9 vs 57.1; OFT-long 98 vs released 94.5; pi0.5 97.6 vs ~96.9; Cosmos3 96 = 96). The accidental default nu = 2d underperforms only where the hard-but-learnable residual band matters (OFT-long 82, Cosmos3 93). The fixed prescription c = 2 (nu = 4d, the same rule on every stack) matches the swept optimum where it has been run to its endpoint (WidowX 67.4 = sweep best; Cosmos3 96 = flow; GR1 51.5 > the swept-at-2d incumbent 47.5, single seed, SE ~2.3), is within 3 points on OFT-long (95 vs 98 for nu=1024 at 150k; above the released L1 94.5), and is 1.0 point below the 2d incumbent on pi0.5 (96.6 vs 97.6 over 4000 episodes, single seed; the gap sits in libero-long, 92.6 vs 94.7, the other three suites are within 0.7). pi0.5 is the one stack where the lighter tail does not help.

Huber-gate controls (budget-matched, not a table column): GR1 simple Huber (MSE inside 2.5 x EMA-median chunk-residual norm, linear beyond; no sigma) 37.4 @60k = MSE 37.8, vs HT 47.5 / flow 44.1; GR1 heteroscedastic Huber (Huber rho on ||r||/sigma + d log sigma) 40.7 @60k: sigma normalization recovers about half of the MSE->HT gap, the rescending gate the rest. OFT-long simple / hetero Huber 97 / 95 @50k (arms stopped at 80k). The cap reproduces HT on OFT-long but not on GR1.

Budget-mismatched controls (not in the table): OFT-long hetero-Gaussian 73 @50k (arm stopped at ~91k; 59 @20k); OFT-goal/object HG 94/88 @50k; GR1 pure HG 20.6 @22k and HG-pre->HT 32.8 @22k (HT @60k); in-house OFT L1 66 @20k and MSE 84 @20k. GOOGLE ROBOT (FRACTAL) COMPLETED 2026-09-02, and it is the FIRST STACK WHERE HT LOSES TO FLOW: flow 67.3 vs HT nu=224 63.5 (6 tasks x 100 eps, identical recipe/data/eval, full 87,212-episode dataset, sbias -0.5093). Per task flow/HT: coke 0.88/0.94, pick-object 0.76/0.82, move-near 0.99/0.94, open-drawer 0.59/0.56, close-drawer 0.75/0.50, place-in-closed-drawer 0.07/0.05 (released N1.7: 1.00/0.94/1.00/0.65/0.69/0.07 = 72.5). The split is systematic: HT WINS both free-space pick tasks (+6 each) and LOSES the articulated/contact tasks (close-drawer -25). Both arms sit below the released reference (flow -5, HT -9), so the absolute level is set by our training rather than by the head. Contrast WidowX (same d=56, same c=2): HT 67.4 vs flow 57.1. Single seed, 100 eps/task (SE ~5 pts per task, ~2 pts on the mean). LIBERO-LONG HAS A PER-TASK TRAINING BIFURCATION (2026-09-01) -- single-run comparisons on that suite are unsafe. Matched 500-episode evals of the long arms: nu=350 (c=2.5) gives 80.0 @50k and 81.2 @100k vs nu=224 (c=2) 93.2 / 92.2 (and 90.2 @20k, 94.0 @150k). The whole 12-point gap sits in TWO of the ten tasks (task 1: 0.18-0.20 vs 0.88-0.94; task 2: 0.60-0.68 vs 0.96-0.98); on the other eight tasks the arms are indistinguishable (91.5 vs 92.0). A smooth nu effect would degrade all ten tasks slightly, so this is a run-level failure to learn specific long-horizon tasks, not a nu effect -- and it explains the earlier non-monotone reading (nu=112 82, nu=224 93, nu=350 80, nu=1024 93: two 'bad' runs and two 'good' ones). Quote libero-long only with several seeds, or report the eight-task subset alongside. NU SWEEP ON OFT IS FLAT ONCE CONFIRMED AT 500 EPISODES (2026-08-31). The swept-best nu=1024 row was built from 100-episode screens; re-run at 500 eps it gives long 93.0 (screen 98), goal 97.6 @150k / 96.0 @50k (screen 98), object 95.8 @20k (screen 99) -- i.e. nu=1024 is NOT better than the fixed nu=224 (long 94.0, goal 96.6/96.8, object 95.6/96.6); on the three suites with both, the means are 95.5 (nu=1024) vs 95.4-96.0 (nu=224). OFT therefore does NOT prefer a large c: its apparent optimum at c=4.28 was screen noise, and no value of nu closes the ~1-point gap to the released L1 (97.1). This removes the only stack that contradicted the fixed c=2 prescription. C=2.5 SWEEP (nu=6.25d), started 2026-08-30 after the c=2 column closed: WidowX nu=350 65.1 (per task 0.54/0.82/0.90/0.82/0.44/0.86/0.18) vs c=2 67.4 and 2d 62.9 -- same within the pooled SE of ~6 points (5 parallel envs share initial states, so 10 distinct scenes/task). GR1 nu=1450 47.6 @60k (24 tasks x 20 eps, SE 2.3) vs c=2 51.5 and 2d 47.5: going heavier than 4d does not help on GR1. pi0.5 nu=2187.5 97.2 (spatial 97.6 / goal 97.0 / object 99.6 / long 94.5, 4 x 1000 eps, SE 0.35 on the average) vs c=2 96.6 and 2d 97.6: NOT monotone -- the c=2 point is the low one, driven by libero-long (92.6 vs 94.5-94.7 at c=1.41 and c=2.5), so pi0.5 is flat over c=1.41..2.5 and both neighbours of c=2 sit above the published flow baseline (96.9). OFT-long nu=350 and OFT-object nu=350 still training. Arm status (2026-08-27). Completed: OFT-long fixed nu=224 (89 @20k, 94 @30k, 94 @50k, 95 @150k), Cosmos3 fixed nu=640 (62/83/97/96 @iter 500/1000/1500/2000), GR1 simple / hetero Huber (37.4 / 40.7 @60k), OFT-long learned-nu (93 @150k). Stopped before their endpoint (last evaluated snapshot is final): OFT-long median-pinned c=2.5 (95 @50k, stopped at 130k) and c=2 (88 @50k, at 60k), per-sample / global argmax-nu table (92 / 91 @50k, at 80k), OFT-long simple / hetero Huber (97 / 95 @50k, at 80k), Cosmos3 median-pinned c=2 (never evaluated). Completed for the c = 2 column: GR1 nu=928 51.5 @60k, pi0.5 nu=1400 96.6 @30k (8-GPU rerun, budget-matched; a first 1-GPU attempt gave 96.1). Completed 2026-08-30, all OFT c=2 cells re-run at 500 episodes: spatial 97.4 (487/500), object 95.6 (478/500), goal 96.6 @150k (483/500) and 96.8 @50k (484/500), long 94.0 (470/500) -> 4-suite average 96.0 (paper protocol: 150k, goal 50k) vs released L1 97.1; 96.2 if object also uses its selected checkpoint (96.6 @100k). The 100-episode screens read high on every suite (goal 99 -> 96.6, long 95 -> 94.0, object 94 -> 95.6 low), consistent with the runbook's screen-vs-confirm rule; the c = 2 column is now complete for every stack. Checkpoint protocol: OpenVLA-OFT's own recipe (LIBERO.md) evaluates the 150k checkpoint for every suite except libero-goal, where they use 50k; our runs copy their command including the 10x LR decay at 100k, so all OFT cells above are 150k endpoints unless marked otherwise. A 13-point checkpoint sweep of object nu=224 at 100 episodes (10k 92, 20k 96, 30k 95, 40k 96, 50k 94, 60k 94, 70k 96, 80k 97, 90k 94, 100k 98, 120k 97, 140k 96, 150k 94) is flat within noise (SE 2.2 points at n=100); the first 100-episode reading of 94 at 150k was noise, and the 500-episode re-evaluation gives 95.6. DROID + Cosmos3 flow/HT (nu=1024) pair: data, recipe and smoke test done (4

samples/rank fit; 108 s per 512-sample iteration on one A800 node, i.e. ~6 days for 5k iterations), paused until the c = 2 cells are filled; no RoboLab evaluation stack exists yet. Full trajectories: memory note `nu-is-inert-heteroscedasticity-is-the-method.md` and `vla_results.md`.

B. Single-pass vs iterative cost

- HT is 1 NFE on every stack. Iterative baselines: GR1 flow 4 NFE, pi0.5 flow 10, Cosmos3 30. Cosmos3's HT is served through the same 30-step joint video/action sampler, so it is not a latency win there.
- OpenVLA-OFT's diffusion head at $T_{\text{test}} = 1$ collapses to 0.0 SR (their Table II): single-pass by construction is not free for iterative heads.

C. Mechanism results (mechanism section)

- Gradient share of the top-1% residual samples: HG 18.5% vs HT (nu=1024) 0.9% on OFT (per-sample sigma); pi0.5 9.4% -> 1.0%. Data/gradient proportion tables in the memory note.
- GR1 HG-vs-HT training-set fitting probe: HG fits worse on every channel; worst on the binary hand-transition channels (HG/HT rms ratio 1.44 vs 1.25 elsewhere, dim 24 = 1.89).
- Multimodality: 0 alternative-action states in 3000 screened states per stack; all above-chance sampling structure is gripper/hand event timing (`fig_flow_unimodal.png`, `multimodality_table.md`).
- Flow vs HG vs HT over the surprise tail (E7; same samples, flow checkpoints): HG AMPLIFIES the tail, flow is PROPORTIONAL to it (gradient share ~ data share; the (t, noise) draw dominates per-sample gradient, corr with the sample's residual ~0), HT SUPPRESSES it:

stack	tail bucket	data %	HG grad %	flow grad %	HT grad %
GR1 (flow ckpt @60k)	1-2x rms	23.4	40.0	28.0	27.0
GR1	2-4x rms	0.6	2.8	1.2	0.6
GR1	top 1% of samples	1.0	4.0	2.9	1.1
pi0.5 (flow ckpt libero_ft)	1-2x rms	36.0	62.4	36.6	53.3
pi0.5	2-4x rms	3.0	15.8	2.4	3.7
pi0.5	top 1% of samples	1.0	6.2	4.0	1.5

Matches the SR ordering HT >= flow > HG (GR1 47.5 / 44.1 / 20.6; OFT-long 98 / -- / 73).

- ML nu on converged residuals is 2.0-24 on every stack while SR-optimal nu is 224-1024: likelihood-optimal and learning-optimal gates are different objects.

D. Caveats to carry into the paper

- OFT cells: 100 episodes, single seed; the released reference is best-of-checkpoints over 500 eps.
- WidowX: 5 parallel envs share initial states -> effective n = 10 scenes per task (SE ~5.9 pts pooled).
- GR1 HG (22k) vs HT (60k): HG stopped early; the per-dimension structure of the fit gap is the step-robust signal, the uniform part is partly budget.
- pi0.5 "ref flow ckpt" numbers are our-harness evals of finetuned_v044 and sit well below the published figures; quote the published number as the baseline.
- GR1 HT best full-decay endpoint is 47.4 at a 50k schedule (no matched flow run at 50k); the strictly matched comparison is the 60k endpoint.

E. Gradient proportion tables (why HT > HG at large nu)

Method: per-sample residuals from several hundred training samples at each stack's trained HT checkpoint (dumps `resid_dist_{oft,pi05,gr1}.npz`; OFT = HTpair-long 120k, pi0.5 = run_ht 30k, GR1 = ft_ht3 60k). "data %" = share of samples in the bucket (error size in multiples of the pooled rms); "grad %" = the bucket's share of total squared-gradient mass, HG mass = S/σ^4 , HT mass = $w^2 * S/\sigma^4$ with $w = (nu+d)/(nu + S/\sigma^2)$. Measured at converged, gate-trained checkpoints, so the tails are as small as they ever get: these are lower bounds on what HG suffers during training.

E1. OpenVLA-OFT libero-long (960 samples, d=56, nu=1024, per-sample sigma = the loss's actual gate)

error size	data %	HG grad %	HT grad %
< 0.5x rms	46.0	25.7	44.0
0.5-2x (typical)	49.2	51.2	53.9
2-4x	4.0	17.3	1.8
4-10x	0.8	5.9	0.3
> 10x	0.0	0.0	0.0
top 1% of samples	1.0	18.5	0.9

E2. pi0.5 LIBERO (600 samples, d=350, nu=700, pooled sigma)

error size	data %	HG grad %	HT grad %
< 0.5x	24.5	2.9	7.7
0.5-2x	72.8	79.6	89.3
2-4x	2.7	17.5	3.1
4-10x	0.0	0.0	0.0
top 1%	1.0	9.4	1.0

E3. GR00T GR1 robocasa (320 samples, d=232, nu=464, pooled sigma)

error size	data %	HG grad %	HT grad %
< 0.5x	0.0	0.0	0.0
0.5-2x	99.4	97.4	99.4
2-4x	0.6	2.7	0.6
top 1%	1.0	3.9	1.0

GR1 note: chunk-level ratios are compressed by averaging over 232 dims (max 2.1x rms), so at pooled sigma the losses barely differ; GR1's spreading happens through its per-sample sigma head (median 0.035 vs pooled rms 0.199; 97.5% of samples past the gate knee under the trained sigma). The per-sample-sigma version would look like E1.

E4. Gradient pull per sample vs error size (OFT, sigma co-adapted to each nu; typical sample = 1)

loss	sigma / rms	gate knee	0.5x	1x	2x	3x	5x	10x
HG (nu=inf)	1.00	--	0.50	1.00	2.00	3.00	5.00	10.00
nu=1024 (swept best)	0.87	3.7x rms	0.53	1.00	1.66	1.94	1.90	1.29
nu=112 (incumbent)	0.63	0.9x rms	0.86	1.00	0.75	0.55	0.35	0.18
nu=2.2 (ML on residuals)	0.45	0.09x rms	1.95	1.00	0.50	0.34	0.20	0.10

Pull = gate weight x residual. Large nu keeps the rising (MSE-like) pull through the learnable 2-4x band and caps only the far tail; heavy nu inverts the curve (most pull to already-fit samples); HG's pull grows without bound.

E5. ML-optimal nu on the same converged residuals (the objective-mismatch result)

stack	ML nu (one fitted sigma)	ML nu (head's per-sample sigma)	SR-optimal nu	frac past knee at ML nu / at SR-optimal nu
OFT (d=56)	2.2	2.2	1024	100% / 1.3%
pi0.5 (d=350)	2.0	--	700	99.8% / 24%
GR1 (d=232)	23.6	--	464	100% / 3.7%

Maximum likelihood in nu is a fit criterion (it explains the over-dispersed normalized residuals with tails, halving sigma); the SR-optimal nu is a control criterion (keep MSE pull through the learnable band, gate only surprises). A likelihood-fitted nu therefore lands heavy.

E6. Gate scale: theory vs real training, with the swept optimum per stack

Knee in sigma units = $\sqrt{\text{nu}/d}$. Theory row assumes a calibrated sigma and Gaussian residuals ($u \sim \chi^2_d$). "past knee" = measured fraction of training samples with $S/\sigma^2 > \text{nu}$ at the listed checkpoint (per-sample head sigma for OFT/GR1/WidowX, pooled sigma for pi0.5).

stack (d)	setting	nu	k = nu/d	nominal knee	past knee (theory, Gaussian)	past knee (measured)
OFT (56)	incumbent 2d	112	2	1.41 sigma	~1e-5	61%
OFT	fixed c=2 (4d)	224	4	2.0 sigma	6e-22	29%
OFT	swept best	1024	18.3	4.3 sigma	~0	3.6%
GR1 (232)	incumbent 2d = swept best	464	2	1.41 sigma	~1e-12	97.5% (sigma under- predicts 5.5x; Cauchy- like regime)
GR1	fixed c=2 (4d)	928	4	2.0 sigma	6e-84	90%
WidowX (56)	incumbent 2d	112	2	1.41 sigma	~1e-5	12.8% (gate probe @20k)
WidowX	swept best (flat axis)	224	4	2.0 sigma	6e-22	not measured
pi0.5 (350)	incumbent 2d = swept-equivalent (flat)	700	2	1.41 sigma	~1e-19	24% (pooled sigma)
Cosmos3 (160)	incumbent 2d	320	2	1.41 sigma	~1e-9	not measured
Cosmos3	swept best	512	3.2	1.79 sigma	~1e-24	not measured
Cosmos3	fixed c=2 (4d)	640	4	2.0 sigma	1e-58	not measured (SR 96 @2000)

Reading: the swept-optimal k is not stack-invariant (2 to 18, a 9x spread) because sigma's calibration is not; in realized terms the SR-optimal gate touches a few percent of samples on OFT but ~all samples on GR1 (whose optimum operates as a Cauchy-like loss).

E7. Flow-matching gradient distribution (does flow's training gradient concentrate on the tail?)

Method: per training sample, the flow-matching loss is measured with 8 Monte-Carlo draws of (t, noise) on the FLOW checkpoint, and the sample's action-space residual comes from the flow model's own sampled action (eval-setting sampler) vs ground truth. Flow's loss is plain MSE on the velocity residual, so a sample's gradient mass is proportional to its per-sample flow loss. HG/HT columns are computed on the same samples' action residuals (HG mass $\propto S$; HT mass $\propto w^2 S$ at the stack's incumbent nu, pooled sigma).

E8a. GR00T GR1 robocasa (320 samples, ft_flow @60k, 4-step sampler; d=232)

bucket (flow model's action residual)	data %	FLOW grad %	HG grad %	HT (nu=464) grad %
0.5-1x rms	75.9	70.7	57.2	72.4
1-2x	23.4	28.0	40.0	27.0
2-4x	0.6	1.2	2.8	0.6
top 1% of samples	1.0	2.9	4.0	1.1

Diagnostics: corr(per-sample flow loss, action residual) = 0.25; per-sample flow loss varies across the 8 (t, noise) draws with median CV = 0.95 — i.e., most of a sample's flow gradient is set by the random (t, noise) draw rather than by the sample's own residual. Flow's gradient distribution on GR1 sits between HG and HT: it does not concentrate on the tail like HG (the t/noise averaging dilutes per-sample differences) but it also does not gate; note GR1's chunk-level residuals have no far tail at pooled sigma (max ~2.1x rms), so this is the mild-tail case.

E8a-T. The same table across training (GR00T GR1, 320 samples, 8 (t, noise) draws, 4-step sampler; buckets relative to each checkpoint's own pooled rms)

checkpoint	pooled rms	bucket	data %	FLOW grad %	HG grad %	HT (nu=464) grad %
2k (ft_flow2k, SR 1.5%)	0.276	0.5-1x	69.7	62.0	40.8	61.9
		1-2x	30.0	37.7	57.9	37.7
		2-4x	0.3	0.2	1.3	0.4
		top 1% of samples	1.2	1.6	4.6	1.5
8k (ft_flow2k)	0.261	0.5-1x	70.6	65.7	44.9	64.1
		1-2x	29.1	33.5	53.6	35.6
		2-4x	0.3	0.7	1.5	0.3
		top 1% of samples	1.2	1.9	4.5	1.4
46k (keep/flow_46000)	0.217	0.5-1x	75.6	70.3	56.3	71.9
		1-2x	23.8	28.5	40.8	27.4
		2-4x	0.6	1.2	2.9	0.6
		top 1% of samples	1.2	2.1	5.2	1.3
60k (ft_flow, converged; = E8a)	0.214	0.5-1x	75.9	70.7	57.2	72.4
		1-2x	23.4	28.0	40.0	27.0
		2-4x	0.6	1.2	2.8	0.6
		top 1% of samples	1.2	1.9	5.2	1.3

Diagnostics per checkpoint: 2k (ft_flow2k, SR 1.5%): rms 0.276, max residual 2.03x rms, corr(flow loss, residual) 0.46, median CV of flow loss across draws 0.66, flow loss median 0.1725; 8k (ft_flow2k): rms 0.261, max residual 2.18x rms, corr(flow loss, residual) 0.31, median CV of flow loss across draws 0.76, flow loss median 0.1885; 46k (keep/flow_46000): rms 0.217, max residual 2.17x rms, corr(flow loss, residual) 0.27, median CV of flow loss across draws 0.94, flow loss median 0.1939; 60k (ft_flow, converged; = E8a): rms 0.214, max residual 2.11x rms, corr(flow loss, residual) 0.27, median CV of flow loss across draws 0.95, flow loss median 0.1929.

Reading: the three regimes are already present at 2k and do not change in kind with training - HG's share of the 1-2x band and of the top 1% is above the data share at every stage, HT's is at or below it, flow's sits between (flow's top-1% share stays within 1-3x its data share at every stage, HG's within 2-5x). What changes with training is the residual scale (pooled rms falls) and the tail's extent; the ordering HG > flow > HT on the tail is stage-independent. No GR1 flow checkpoint exists between 8k and 46k (the ft_flow run kept only its last five), so the mid-early part of the curve is not covered; pi0.5 has no early/mid flow checkpoints at all (the flow baseline is the released endpoint).

E8b. pi0.5 LIBERO (600 samples, libero_ft flow ckpt, Beta-sampled t, 8 (t, noise) draws; 10-step sampler for the action residual; d=350)

bucket (flow model's action residual)	data %	FLOW grad %	HG grad %	HT (nu=700) grad %
< 0.5x rms	26.2	28.1	3.6	9.5
0.5-1x	34.8	32.9	18.2	33.5
1-2x	36.0	36.6	62.4	53.3
2-4x	3.0	2.4	15.8	3.7
top 1% of samples	1.0	4.0 (single draw 5.8)	6.2	1.5

Diagnostics: corr(per-sample flow loss, action residual) = -0.03 (none); per-sample flow loss CV across draws = 0.63. The flow gradient distribution tracks the DATA distribution almost exactly (28/33/37/2.4 vs 26/35/36/3.0): flow's per-sample gradient is set by the random (t, noise) draw and is essentially blind to whether the sample is a surprise. It neither chases the tail (HG puts 78% of its gradient on the >1x band, 16% on the 2-4x surprise

band) nor gates it (HT: 4%); the tail's share under flow equals its data share. Probe pitfall (recorded): the residual scaffold's `sample_time` -> ones override must be removed for a flow-loss measurement — with $t=1$ the loss collapses to the single-pass residual and all draws coincide.

Summary of E7: three distinct gradient regimes over the surprise tail — HG amplifies it (share $\gg$ data share), flow is proportional to it (share $\sim$ data share; the (t, noise) noise floor dominates per-sample differences), HT suppresses it (share $\ll$ data share). This is consistent with the success-rate ordering $HT \geq \text{flow} > HG$ on GR1 and OFT-long.